Instituto de Física da USP

Report

Nara Avila Moraes

Physics Institute
IFUSP

2026 / 07 / 03

Summary

General Tasks Status

Task	Status	
2026-06 RETINHA presentation	✓	Link
2026-06 Scientific report	◑	Link

Task	Status	
Journal Club - Automations	◑	

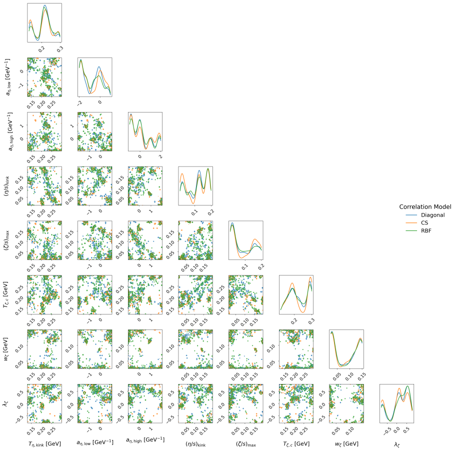
Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
↪ Mcmc Params	✓	Plots
↪ RDF Model	✓	Plots
↪ CS Model	✓	Plots
↪ None Uncertainty Correlation	✓	Plots
Atlas ins1759506 studies		
↪ Uncertainty correlation	◔	
RHIC Au-Au/d-Au Studies JETSCAPE		
↪ Uncertainty correlation	◔	

Baseline vs all correlation model

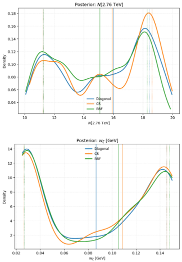
Figure

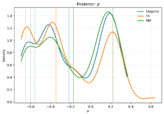
Baseline vs all correlation model



Figure

Preliminary results





14/17
parameters with
$\Delta < 0.3\sigma$

3/17
parameters with
$\Delta > 0.3\sigma$

Changing the experimental covariance structure (Diagonal $\rightarrow$ CS $\rightarrow$ RBF) produces no statistically significant change in the inferred posterior for the Universal Scaled Spectra calibration, with posterior shifts remaining below 0.6σ across all inferred parameters.

Largest posterior shifts

Parameter	Max shift
p	0.515σ
w_ζ	0.418σ
N	0.307σ

Largest normalized median shifts among the 17 inferred parameters.

Posterior comparison for ALICE Pb–Pb $\sqrt{s_{NN}} = 2.76$ TeV, universal spectra calibration.

Figure

Selected Article

[Submitted on 30 Jun 2026]

Relativistic magnetohydrodynamics from kinetic theory

Khwahish Kushwah

This thesis develops a kinetic-theory framework for relativistic dissipative magnetohydrodynamics under strong electromagnetic fields, motivated by quark-gluon plasma in heavy-ion collisions. Starting from the relativistic Boltzmann-Vlasov equation and using the method of moments within the 14-moment approximation, it derives causal second-order hydrodynamic equations for relativistic plasmas with increasing generality. The work first review relativistic dissipative hydrodynamics and its kinetic foundations, emphasizing the need for Israel-Stewart-type transient theories to preserve causality and stability. Electromagnetic fields are then introduced at the microscopic level, where the Lorentz force modifies the moment hierarchy and produces anisotropic transport effects absent in field-free fluids. Next, it develops relativistic dissipative magnetohydrodynamics for a non-resistive two-component plasma of oppositely charged particles. Here, the magnetic field couples the dissipative sectors of the two species, generating relative dissipative currents and coupled shear dynamics. For Bjorken expansion, the theory predicts damped oscillations in the transverse shear sector associated with cyclotron motion. Finally, the thesis treats the resistive two-component case, where the electric field evolves dynamically and couples to charge diffusion and shear stress. The resulting theory reveals current-shear feedback, transient electromagnetic generation of momentum anisotropy, and underdamped dissipative oscillations. Applications to homogeneous and Bjorken-expanding plasmas show how resistive and electromagnetic effects modify evolution beyond standard hydrodynamics. Overall, the thesis extends relativistic dissipative hydrodynamics to magnetized and resistive plasmas, providing a microscopic foundation for future studies of strongly magnetized quark-gluon plasma and astrophysical systems.

Comments: PhD thesis
Subjects: **Nuclear Theory (nucl-th)**; High Energy Astrophysical Phenomena (astro-ph.HE); High Energy Physics - Theory (hep-th); Plasma Physics (physics.plasm-ph)

Figure: PhD Thesis on Magnetohydrodynamics